arc-agi-small-language-model

Deep Learning │ ├── 1. Modelos Perceptivos │ ├── CNNs │ ├── Vision Transformers (ViT) │ ├── Multimodal Encoders │ ├── 2. Modelos Generativos │ ├── GANs │ ├── VAEs │ ├── Diffusion Models │ ├── Flow-based Models │ ├── 3. Modelos Secuenciales │ ├── RNN / LSTM / GRU │ ├── Transformers │ ├── State Space Models (Mamba, S4) │ ├── 4. Foundation Models (Base LLMs) │ ├── LLMs │ │ ├── GPT-3 ──────────────────── ARC-AGI-1: ~0% │ │ ├── GPT-4 ──────────────────── ARC-AGI-1: ~0% │ │ ├── GPT-4o ─────────────────── ARC-AGI-1: 5% │ │ ├── GPT-4.1-Nano ────────────── ARC-AGI-1: 0% | $0.002/task │ │ ├── Llama 4 Maverick ───────── ARC-AGI-1: 4.4% | $0.0078/task | ARC-AGI-2: 0% │ │ ├── Grok 3 ─────────────────── ARC-AGI-1: 5.5% | $0.0093/task | ARC-AGI-2: 0% │ │ └── Claude 3.5 Sonnet ──────── ARC-AGI-1: 14% (base, sin técnicas adicionales) │ ├── Vision Foundation Models │ └── Multimodal Foundation Models │ ├── 5. Neuro-Symbolic & Reasoning AI │ ├── Neuro-symbolic models │ ├── Program Induction Models │ │ ├── Test-Time Training (MIT/Cornell) ── ARC-AGI-1: 47.5% │ │ └── CompressARC (76K params) ───────── ARC-AGI-1: 20-34% | ARC-AGI-2: 4% (sin pretraining) │ ├── Neural Theorem Provers │ └── Differentiable Reasoning Systems │ ├── 6. World Models & Cognitive Models │ ├── World Models │ ├── Predictive Processing Models │ ├── Active Inference Models │ └── Cognitive Architectures │ ├── 7. **Reasoning-Centric Models (LRMs - Large Reasoning Models)** │ │ │ ├── TRM (Token Reasoning Models) - Single Chain-of-Thought │ │ ├── OpenAI o1 (low) ───────── ARC-AGI-1: 20.5% | $0.43/task │ │ ├── OpenAI o1 (medium) ────── ARC-AGI-1: 31% | $0.79/task │ │ ├── OpenAI o1 (high) ──────── ARC-AGI-1: 35% | $1.31/task │ │ ├── DeepSeek R1-Zero ──────── ARC-AGI-1: 14% | $0.11/task (sin SFT, solo RL) │ │ ├── DeepSeek R1 ───────────── ARC-AGI-1: 15.8% | $0.06/task | ARC-AGI-2: 1.3% │ │ ├── DeepSeek R1 (5/28/25) ─── ARC-AGI-1: 21.2% | $0.046/task (actualizado) │ │ ├── OpenAI o3 (low) ───────── ARC-AGI-1: 41% | $1.22/task | ARC-AGI-2: 1.9% │ │ ├── OpenAI o3 (medium) ────── ARC-AGI-1: 53% | $2.52/task | ARC-AGI-2: 2.9% │ │ ├── OpenAI o3 (high) ──────── ARC-AGI-1: 60.8% | $0.50/task | ARC-AGI-2: 6.5% │ │ ├── OpenAI o3-pro (high) ──── ARC-AGI-1: 59.3% | $4.16/task | ARC-AGI-2: 4.9% │ │ ├── OpenAI o4-mini (low) ──── ARC-AGI-1: 21% | $0.05/task | ARC-AGI-2: 1.6% │ │ ├── OpenAI o4-mini (medium) ── ARC-AGI-1: 42% | $0.23/task | ARC-AGI-2: 2.3% │ │ ├── OpenAI o4-mini (high) ──── ARC-AGI-1: 58.7% | $0.406/task | ARC-AGI-2: 6.1% │ │ ├── Grok 3 Mini (low) ──────── ARC-AGI-1: 16.5% | $0.0099/task | ARC-AGI-2: 0.4% │ │ ├── Claude Sonnet 4 (16k) ──── ARC-AGI-1: 40% | $0.366/task | ARC-AGI-2: 5.9% │ │ ├── Claude Opus 4 (16k) ────── ARC-AGI-1: 35.7% | $1.25/task | ARC-AGI-2: 8.6% │ │ ├── Claude Opus 4.5 (64k) ──── ARC-AGI-2: 37.6% | $2.20/task | SOTA comercial verificado │ │ ├── Gemini 2.5 Flash ───────── ARC-AGI-1: 33.3% | $0.037/task (mejor balance) │ │ ├── Gemini 2.5 Pro ─────────── ARC-AGI-1: 33% | $0.569/task | ARC-AGI-2: 3.8% │ │ └── Gemini 3 Pro ───────────── ARC-AGI-2: 31% | $0.81/task (baseline) │ │ │ ├── Search-Augmented Reasoning Models (Test-Time Compute Scaling) │ │ ├── OpenAI o3-preview (low) ── ARC-AGI-1: 75.7% | $200/task | 33.5M tokens │ │ ├── OpenAI o3-preview (high) ─ ARC-AGI-1: 87.5% | $4,560/task | 5.7B tokens (172x compute) │ │ └── Claude 3.5 Sonnet + Evolutionary TTC ── ARC-AGI-1: 53.6% (Jeremy Berman) │ │ │ ├── **Tiny Recursive Models (TRM) - Redes Pequeñas con Razonamiento Recursivo** 🆕 │ │ ├── TRM (7M params) ────────── ARC-AGI-1: 45% | ARC-AGI-2: 8% | Paper Award 1st 2025 │ │ │ └── Solo 2 capas, <0.01% parámetros de LLMs, supera DeepSeek R1, o3-mini, Gemini 2.5 Pro │ │ └── HRM (27M params) ───────── ARC-AGI-1: ~40% | Hierarchical Reasoning Model (precursor) │ │ │ └── HRM (Hierarchical Reasoning Models) │ ├── Hierarchical planners │ ├── Modular reasoning networks │ └── Multi-level cognitive models │ ├── 8. Agentic AI Systems │ ├── Autonomous Agents │ ├── Tool-using Models │ │ └── SLM Fine-tuned (OPT-350M) ── ToolBench: 77.55% (supera ChatGPT-CoT 26%) │ ├── Planning Agents │ └── Multi-agent Systems │ ├── 9. **Soluciones Híbridas / Especializadas (Kaggle ARC Prize)** │ │ │ ├── ARC Prize 2024 (ARC-AGI-1) │ │ ├── the ARChitects (1º) ────── 53.5% (Program synthesis + search) │ │ ├── Guillermo Barbadillo (2º)── 40% │ │ ├── alijs (3º) ─────────────── 40% │ │ ├── William Wu (4º) ────────── 37% │ │ └── Ensemble Solutions ─────── ~81% (combinación de múltiples soluciones) │ │ │ └── ARC Prize 2025 (ARC-AGI-2) │ ├── NVARC (1º) ─────────────── 24.03% | $0.20/task | SOTA Kaggle │ ├── the ARChitects (2º) ────── 16.53% | 2D-aware masked-diffusion LLM │ ├── MindsAI (3º) ───────────── 12.64% | Test-time training pipeline │ ├── Lonnie (4º) ────────────── 6.67% │ └── G. Barbadillo (5º) ─────── 6.53% │ └── 10. **Model Refinement Solutions (Refinement Loops)** ├── Poetiq (Gemini 3 Pro Ref) ──── ARC-AGI-2: 54% | $31/task | SOTA refinement ├── Poetiq (Claude Opus 4.5 Ref)── ARC-AGI-2: ~54% | $60/task ├── SOAR (Self-Improving LLM) ──── ARC-AGI-1: 52% | Paper Award 2nd 2025 └── Evolutionary Program Synthesis (E. Pang) ── Paper Award Runner-up

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Resumen de Benchmarks ARC-AGI (Actualizado Dic 2025)

ARC-AGI-1 Semi-Private Eval (Ranking por rendimiento)

Modelo/Sistema	Score	Costo/Tarea	Notas
o3-preview (high)	87.5%	$4,560	172x compute, 5.7B tokens
Ensemble Kaggle	~81%	-	Combinación de soluciones
o3-preview (low)	75.7%	$200	Primer puesto leaderboard 2024
o3 (high)	60.8%	$0.50	-
o3-pro (high)	59.3%	$4.16	-
o4-mini (high)	58.7%	$0.406	-
Claude 3.5 + EvoTTC	53.6%	-	Evolutionary Test-Time Compute
the ARChitects (2024)	53.5%	-	Ganador Kaggle ARC Prize 2024
o3 (medium)	53%	$2.52	-
SOAR	52%	-	Self-Improving Evolutionary Program Synthesis
TTT MIT/Cornell	47.5%	-	Test-Time Training
TRM (7M params)	45%	-	🆕 Tiny Recursive Model - Paper Award 1st
o3 (low)	41%	$1.22	-
Claude Sonnet 4 (16k)	40%	$0.366	-

Modelo/Sistema	Score	Costo/Tarea	Notas
o4-mini (medium)	42%	$0.23	-
Claude Opus 4 (16k)	35.7%	$1.25	-
o1 (high)	35%	$1.31	-
Gemini 2.5 Flash	33.3%	$0.037	Mejor balance costo/rendimiento
Gemini 2.5 Pro	33%	$0.569	-
o1 (medium)	31%	$0.79	-
DeepSeek R1 (5/28)	21.2%	$0.046	Actualizado
o4-mini (low)	21%	$0.05	-
o1 (low)	20.5%	$0.43	-
CompressARC (76K)	20-34%	-	Sin pretraining, MDL-based
Grok 3 Mini (low)	16.5%	$0.0099	-
DeepSeek R1	15.8%	$0.06	Open source
DeepSeek R1-Zero	14%	$0.11	Sin SFT, solo RL
Claude 3.5 Sonnet	14%	-	Base, sin técnicas adicionales
Grok 3	5.5%	$0.0093	-
GPT-4o	5%	-	Base LLM
Llama 4 Maverick	4.4%	$0.0078	-
GPT-4	~0%	-	Base LLM
GPT-3	0%	-	Base LLM
Humanos	~85%	~$5	Referencia humana

ARC-AGI-2 (Benchmark más difícil - lanzado Marzo 2025)

Modelo/Sistema	Score	Costo/Tarea	Notas
Poetiq (Gemini 3 Pro Ref)	54%	$31	🆕 SOTA refinement solution
Claude Opus 4.5 (64k)	37.6%	$2.20	🆕 SOTA comercial verificado
Gemini 3 Pro	31%	$0.81	Baseline
NVARC (Kaggle 1st)	24.03%	$0.20	🆕 SOTA Kaggle 2025
the ARChitects	16.53%	-	2D-aware masked-diffusion
MindsAI	12.64%	-	Test-time training
Claude Opus 4 (16k)	8.6%	$1.93	-
TRM (7M params)	8%	-	🆕 Tiny Recursive Model
o3 (high)	6.5%	$0.834	-
o4-mini (high)	6.1%	$0.856	-
Claude Sonnet 4 (16k)	5.9%	$0.486	-
o3-pro (high)	4.9%	$7.55	-
CompressARC (76K)	4%	-	Sin pretraining
Gemini 2.5 Pro	3.8%	$0.813	-
o3 (medium)	2.9%	-	-
o4-mini (medium)	2.3%	-	-
o3 (low)	1.9%	-	-
o4-mini (low)	1.6%	-	-
DeepSeek R1	1.3%	$0.08	-
Grok 3 Mini (low)	0.4%	$0.013	-
Grok 3	0%	$0.14	-

Modelo/Sistema	Score	Costo/Tarea	Notas
Llama 4 Maverick	0%	$0.012	-
Humanos	>95%	-	Sin entrenamiento previo

Insights Clave (Actualizado 2025)

1. **Los LLMs base fracasan** en ARC-AGI porque dependen de memorización, no razonamiento adaptativo
2. **Los Reasoning Models (o1, R1, o3)** mejoran mediante Chain-of-Thought, pero tienen límites
3. **El test-time compute scaling** permite mejoras logarítmicas pero con rendimientos decrecientes
4. **Tiny Recursive Models (TRM)** logran 45% en ARC-AGI-1 con solo 7M parámetros (<0.01% de LLMs)
5. **ARC-AGI-2 sigue sin resolver**: incluso los mejores modelos <10%, humanos >95%
6. **Refinement Loops** son el tema central de 2025 - mejoran performance iterativamente
7. **La eficiencia importa**: Gemini 2.5 Flash ofrece el mejor ratio costo/rendimiento para ARC-AGI-1
8. **No hay ganador claro**: cada modelo tiene trade-offs entre accuracy, costo y velocidad
9. **ARC-AGI-3** (2026) introducirá razonamiento interactivo: exploración, planificación, memoria

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Bibliografía Relevante

Papers con resultados ARC-AGI

1. **"Less is More: Recursive Reasoning with Tiny Networks"** (Oct 2025)

   - Autor: Alexia Jolicoeur-Martineau
   - arXiv: 2510.04871
   - Resultado: TRM (7M params) → ARC-AGI-1: 45%, ARC-AGI-2: 8%
   - 🏆 Paper Award 1st Place - ARC Prize 2025

2. **"Small Language Models for Efficient Agentic Tool Calling"** (Dec 2025)

   - Autores: Jhandi, Kazi, Subramanian, Sendas
   - arXiv: 2512.15943
   - Resultado: OPT-350M fine-tuned → ToolBench: 77.55% (vs ChatGPT-CoT: 26%)

3. **"On the Measure of Intelligence"** (2019)

   - Autor: François Chollet
   - arXiv: 1911.01547
   - Introduce ARC-AGI benchmark

4. **"The Surprising Effectiveness of Test-Time Training for Abstract Reasoning"** (2024)

   - MIT/Cornell
   - Resultado: ARC-AGI-1: 47.5%